

✓ Suicide Detection - Baseline Model

This notebook builds a baseline using TF-IDF and Logistic Regression.

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import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("Suicide_Detection.csv")
df = df.dropna()
df['label'] = df['class'].map({'non-suicide': 0, 'suicide': 1})

X_train, X_test, y_train, y_test = train_test_split(df['text'], df['label'], test_size=0.2, random_state=42, stratify=df['label'])

vectorizer = TfidfVectorizer(max_features=5000, ngram_range=(1, 2), stop_words='english')
X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)

model = LogisticRegression(max_iter=1000)
model.fit(X_train_vec, y_train)

y_pred = model.predict(X_test_vec)

print(classification_report(y_test, y_pred))
conf_mat = confusion_matrix(y_test, y_pred)

sns.heatmap(conf_mat, annot=True, fmt='d', cmap='Blues', xticklabels=['Non-suicide', 'Suicide'], yticklabels=['Non-suicide', 'Suicide'])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```

